

Wireless Door-Opening Alarm

PRADEEP G.

This wireless door-opening alarm works with ultrasonic waves. Its transmitter is installed near the door and the receiver at some distance. There is no wire connection between the two. When the door is opened, the transmitter emits 40kHz ultrasonic signal which sounds an alarm through the receiver.

Circuit and working

Fig. 1 shows circuit diagram of the transmitter, which employs NE555

timer and CD4013 ICs. NE555 (IC1) timer is configured in astable mode to generate a frequency of 80 kHz. CMOS flip-flop CD4013 (IC2) generates a 40kHz square-wave signal to drive ultrasonic transmitter transducer (TX1).

Reed switch is fitted on the door frame and a magnet on the door such that, when the door is closed, the magnet comes very close to the reed switch, making the reed switch close its contacts. This means, reset pin 4 of IC1 is pulled to ground. NE555 stops oscillation and so there is no output from ultrasonic transducer TX1.

When the door is opened, the magnet moves away from the reed switch and contacts of reed switch open. As a result, NE555 oscillates at 80 kHz and the transducer TX1 emits 40kHz signal.

Transmitter can be powered from a 9V

battery. To reduce current consumption, NE555 may be replaced with 7555 CMOS IC.

Fig. 2 shows the circuit of receiver which uses 40kHz transducer (RX1) at the input. The 40kHz signal reaches the receiver through RX1. Signal from transducer RX1 is amplified, rectified and filtered. DC voltage obtained is given to DC amplifier to drive LED1

Test Points

Test point	Details
TP0	0V
TP1	9V
TP2	80 kHz when reed switch is open
TP3	40 kHz when reed switch is open
TP4	40 kHz when reed switch is open
Across TP5 and TP6	9V -12V

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- IC2 - CD4013 dual D-type flip-flop
- D1 - 1N4148 switching diode
- T1 - BC549 npn transistor
- T2-T4 - BC547 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3 - 5.6-kilo-ohm
- R2 - 18-kilo-ohm
- R4, R8 - 4.7-kilo-ohm
- R5 - 470-kilo-ohm
- R6, R11, R12 - 1-kilo-ohm
- R7, R9 - 22-kilo-ohm
- R10 - 100-kilo-ohm
- R13, R14 - 680-ohm
- VR1 - 20-kilo-ohm preset
- VR2 - 2.2-mega-ohm preset

Capacitors:

- C1 - 330pF ceramic disk
- C2 - 0.01 μ F ceramic disk
- C3 - 100 μ F, 25V electrolytic
- C4 - 0.1 μ F ceramic disk
- C5 - 4.7nF ceramic disk
- C6, C8 - 100 μ F, 25V electrolytic
- C7 - 47nF ceramic disk

Miscellaneous:

- RS1 - Reed switch (N/O type)
- TX1 - 40kHz ultrasonic transducer (transmitter)
- RX1 - 40kHz ultrasonic transducer (receiver)
- CON1, CON2 - 2-pin terminal connector
- PZ1 - Piezo buzzer
- 9V battery

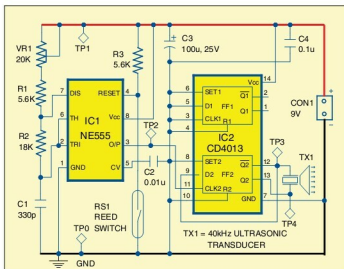


Fig. 1: Circuit diagram of the transmitter

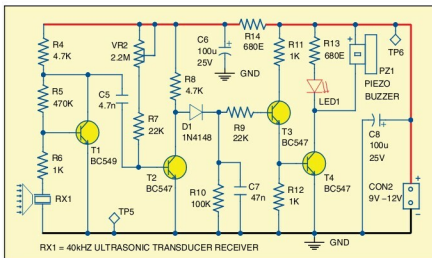


Fig. 2: Circuit diagram of the receiver